

Log-periodic oscillation in the all-heads coin game

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Abstract

We study the large- n behaviour of the all-heads coin game below the fair coin. Two phenomena are established. First, for every $p \in (0, \frac{1}{2})$ the value $b_{n,p}$ of the greedy strategy ALL is asymptotically *log-periodic*: it is asymptotic to a continuous, non-constant function of $\log n$ with period $\log(1/(1-p))$, and in particular does not converge. Second — and this is the main result — the *optimal* winning probability $w_{n,p}$ inherits this oscillation: for every $p \in (0, \frac{1}{2})$ the sequence $n \mapsto w_{n,p}$ is not eventually monotone and has infinitely many strict local maxima and minima. The proof reduces the statement to two one-sided assertions, treats the increasing side by an elementary Bellman estimate, and the decreasing side by an exact transfer to strategy ALL followed by a renormalisation argument that propagates the oscillation across geometric scales. This paper is the asymptotic companion to the exact and perturbative analysis of [1].

*Working draft. The one step still to be completed is flagged in the text by a boxed **OPEN GAP** marker (Gap D); Section 9 consolidates its status.*

1 Introduction

1.1 The game

Fix $p \in (0, 1)$ and write $q := 1 - p$. A player holds n coins, each of which lands heads independently with probability p . In each round all remaining coins are tossed; the player must set aside at least one coin showing heads, and loses if no coin shows heads. The game is won once every coin has been set aside. Writing $w_{n,p}$ for the optimal winning probability, the dynamic-programming principle gives the Bellman equation

$$w_{n,p} = p^n + \sum_{j=1}^{n-1} \binom{n}{j} p^{n-j} q^j \max_{j \leq m \leq n-1} w_{m,p}, \quad w_{0,p} := 1. \quad (1)$$

Two strategies are natural: ONE sets aside a single head each round (value $a_{n,p}$), and ALL sets aside *every* head. The winning probability $b_{n,p}$ of strategy ALL is defined by the linear recursion

$$b_{n,p} = \sum_{j=0}^{n-1} \binom{n}{j} p^{n-j} q^j b_{j,p}, \quad b_{0,p} := 1, \quad (2)$$

obtained by conditioning on the number j of coins showing tails in the first round: those j coins are re-tossed, while the $n - j \geq 1$ heads are set aside for good (the term $j = 0$ has weight p^n and wins immediately). The companion paper [1] determines $w_{n,p}$ *exactly* at $p = \frac{1}{2}$, identifies the limit $W(p) = \lim_n w_{n,p}$ for $p > \frac{1}{2}$, and gives a first-order perturbation expansion for $p < \frac{1}{2}$ near $\frac{1}{2}$. All of those results are machine-checked in Lean 4.

The present paper concerns the regime left open by a perturbative analysis: the genuine large- n behaviour for $p < \frac{1}{2}$, which turns out to be *oscillatory* rather than convergent.

1.2 Main results

We state the two theorems at once; everything else in the paper is in their service.

Theorem 1.1 (oscillation of the optimal value). *For every $p \in (0, \frac{1}{2})$ the sequence $n \mapsto w_{n,p}$ is not eventually monotone. It has infinitely many strict local maxima and infinitely many strict local minima.*

Theorem 1.2 (log-periodicity of strategy ALL). *For every $p \in (0, \frac{1}{2})$ there is a continuous, strictly positive, non-constant function $\Phi_p : \mathbb{R} \rightarrow \mathbb{R}$, periodic with period $L := \log(1/q)$, such that*

$$b_{n,p} = \Phi_p(\log n) (1 + o(1)) \quad (n \rightarrow \infty).$$

In particular $b_{n,p}$ does not converge: $\liminf_n b_{n,p} < \limsup_n b_{n,p}$.

Theorem 1.1 sharpens, and corrects the natural first guess about, the first-order picture of [1]: that analysis shows $w_{n,p}$ to be eventually increasing *at first order in $\frac{1}{2} - p$* , but the exact sequence is not eventually monotone at all. The oscillation is non-perturbative — its amplitude is exponentially small in $1/(\frac{1}{2} - p)$ — which is why it is invisible to any finite-order expansion.

1.3 Strategy of proof

The proof of Theorem 1.1 has a rigid skeleton, which we record now so that the role of each later section is clear. For a real sequence (x_n) put $d_n := x_{n+1} - x_n$.

Step 0 (reduction). A bounded sequence has infinitely many strict local maxima and minima as soon as it is *not eventually monotone*, i.e. as soon as $d_n > 0$ infinitely often and $d_n < 0$ infinitely often (Lemma 5.1). Thus Theorem 1.1 splits into

(A) $(w_{n,p})$ is not eventually non-decreasing, (B) $(w_{n,p})$ is not eventually non-increasing.

Step A (the increasing side). On a stretch where w is non-decreasing, the suffix-maximum in (1) collapses to its right endpoint and the Bellman equation becomes the linear strategy-ONE recursion, whose one-step increment has the sign-definite dominant term $-q^n w_{n-1}$. Since $q > p$, this forces an eventual decrease. Statement (A) follows (Proposition 4.1); the argument is elementary and complete.

Step B (the decreasing side). On a stretch where w is non-increasing, the suffix-maximum collapses to its *left* endpoint, and w becomes — exactly — a strategy-ALL sequence (Lemma 5.2). Strategy ALL oscillates (Theorem 1.2); hence w cannot have a non-increasing tail. The transfer of the oscillation from ALL to w is carried out by a renormalisation argument (Section 6): a windowed fundamental coefficient $A(t)$ is shown to satisfy a propagation identity across the geometric scale, from which non-convergence — hence non-monotonicity — follows.

The asymmetry between Steps A and B is structural: the increasing side reduces to a contractive linear recursion with a sign-definite drift, the decreasing side to a scale-invariant recursion whose solutions oscillate. Step A is elementary; Step B carries the analytic weight of the paper.

2 Setup and notation

Throughout, $p \in (0, \frac{1}{2})$ is fixed, $q = 1 - p \in (\frac{1}{2}, 1)$, and we use freely that $q > p$, $p/q < 1$ and $2p < 1$. We write $L := \log(1/q) > 0$ and $\omega := 2\pi/L$.

The value $w_{n,p}$ satisfies (1); it is a probability, so $0 < w_{n,p} \leq 1$, and in fact $w_{n,p} < 1$ for $n \geq 1$. The two reference strategies are dominated by the optimum. Strategy ALL has winning probability $b_{n,p}$, defined by the recursion (2); strategy ONE satisfies the linear recursion

$$a_{n,p} = p^n + (1 - p^n - q^n) a_{n-1,p}, \quad a_{0,p} := 1. \quad (3)$$

In both cases $w_{n,p} \geq \max(a_{n,p}, b_{n,p})$, since the optimum dominates any fixed strategy. We use the binomial identities $\sum_{j=1}^{n-1} \binom{n}{j} p^{n-j} q^j = 1 - p^n - q^n$ and $\sum_{j=0}^{n-1} \binom{n}{j} p^{n-j} q^j = 1 - q^n$. Where the value of p is fixed we abbreviate $w_n := w_{n,p}$, $b_n := b_{n,p}$.

3 Strategy All: log-periodic asymptotics

This section establishes Theorem 1.2 — the existence of the log-periodic profile Φ_p — save for one point, the non-constancy of Φ_p , which is Gap D (Section 7). The route is elementary: an exact product identity for the generating function, followed by analytic de-Poissonisation. Neither Hayman admissibility theory nor a Mellin contour shift is needed.

3.1 The functional equation

Let $B(z) := \sum_{n \geq 0} b_n z^n / n!$ be the exponential generating function of strategy ALL. The convolution structure of (2) yields the first-order q -difference equation

$$B(z) = 1 + (e^{pz} - 1) B(qz), \quad (4)$$

and iterating it (the remainder tends to 0 since each factor $e^{pq^k z} - 1 \rightarrow 0$),

$$B(z) = \sum_{M \geq 0} \phi_M(z), \quad \phi_M(z) := \prod_{k=0}^{M-1} (e^{pq^k z} - 1). \quad (5)$$

Probabilistically, with $X_1, \dots, X_n$ i.i.d. geometric, $b_{n,p} = \Pr(\{X_1, \dots, X_n\} = \{1, \dots, \max_i X_i\})$.

3.2 An exact product identity

Write $h(z) := \log(1 - e^{-pz})$. From $e^{pq^k z} - 1 = e^{pq^k z}(1 - e^{-pq^k z})$ and $p \sum_{k < M} q^k = 1 - q^M$,

$$\phi_M(z) = e^{z(1-q^M)} \exp S_M(z), \quad S_M(z) := \sum_{k=0}^{M-1} h(q^k z). \quad (6)$$

The associated *convergent* harmonic sum is

$$\tilde{H}(x) := \sum_{j \geq 0} h(q^{-j} x) \quad (7)$$

— here the arguments $q^{-j} x \rightarrow \infty$, so $h(q^{-j} x) \rightarrow 0$ exponentially. Reindexing, $\tilde{H}(q^{M-1} z) = \sum_{k \leq M-1} h(q^k z)$, whence $S_M(z) = \tilde{H}(q^{M-1} z) - \tilde{H}(z/q)$. Substituting this into (6)–(5) and setting

$$\Xi(w) := e^{-w} e^{\tilde{H}(w/q)} \quad (8)$$

yields the exact identity

$$B(z) = e^z e^{-\tilde{H}(z/q)} \sum_{M \geq 0} \Xi(q^M z). \quad (9)$$

3.3 The profile

The function Ξ is positive, smooth, and decays at *both* ends: $\Xi(w) \leq e^{-w}$ as $w \rightarrow \infty$ (since $\tilde{H} \leq 0$), and $\Xi(w) \rightarrow 0$ faster than any power as $w \rightarrow 0^+$, because $\tilde{H}(x) \rightarrow -\infty$ (a sum of $\sim |\log x|/L$ terms each $\approx \log(px)$; in fact $\tilde{H}(x) \sim -(\log x)^2/(2L)$). Hence $\sum_{M \geq 0} \Xi(q^M z)$ is a harmonic sum of a bump. Its bi-infinite completion

$$\Phi(t) := \sum_{M \in \mathbb{Z}} \Xi(q^M e^t) \quad (10)$$

converges locally uniformly; Φ is continuous, strictly positive, bounded, and *exactly* L -periodic. The omitted terms $M \leq -1$ are $O(e^{-cz})$ and $\tilde{H}(z/q) = O(e^{-cz})$, so (9) gives, on the positive axis,

$$B(z) = e^z \Phi(\log z) (1 + O(e^{-cz})), \quad z \rightarrow +\infty. \quad (11)$$

The Fourier coefficients of Φ are $\hat{\Phi}(m) = L^{-1} \mathcal{M}[\Xi](2\pi im/L)$, with $\mathcal{M}[\Xi]$ entire (Ξ decays faster than any power at both ends).

3.4 Transfer to coefficients

The passage from $B(z)$ to $b_n = n! [z^n] B(z)$ is analytic de-Poissonisation [4]. Write $\Psi(z) := B(z)e^{-z}$; the required sector control is read off the product (5).

Lemma 3.1 (sector control). *There are constants C, γ , depending only on p , such that for $z = re^{i\theta}$ with r large,*

$$|B(re^{i\theta})| \leq \begin{cases} C r^\gamma e^{r \cos \theta}, & |\theta| \leq \pi/2, \\ C r^\gamma, & \pi/2 \leq |\theta| \leq \pi. \end{cases}$$

Proof. Let k^* be least with $pq^{k^*}r \leq 1$; then $k^* \leq 1 + \log(pr)/L$. For $k < k^*$ use $|e^{pq^k z} - 1| \leq 1 + e^{pq^k r \cos \theta}$; for $k \geq k^*$ use $|e^w - 1| \leq |w| e^{(\Re w)^+}$, i.e. $|e^{pq^k z} - 1| \leq pq^k r e^{(pq^k r \cos \theta)^+}$. If $\cos \theta \geq 0$, the early factors give $\prod_{k < k^*} 2 e^{pq^k r \cos \theta} \leq 2^{k^*} e^{r \cos \theta}$ (as $\sum_k pq^k = 1$), and the late factors $\prod_{k \geq k^*} e^{pq^k r \cos \theta}$ decay super-exponentially in M ; summing over M a geometric-type series gives the first bound, with $\gamma = \log 2/L$. If $\cos \theta \leq 0$, the early factors are each ≤ 2 and the late ones $\leq pq^k r$, giving the second bound. $\square$

Lemma 3.1 furnishes both de-Poissonisation hypotheses. Inside the cone $|\arg z| \leq \pi/2$, $|\Psi(z)| = |B(z)| e^{-r \cos \theta} \leq C r^\gamma$ is polynomially bounded; outside any cone $|\arg z| \geq \delta$, $|B(z)| \leq C r^\gamma e^{r \cos \delta} \leq e^{\alpha r}$ with $\alpha = \frac{1}{2}(1 + \cos \delta) < 1$. De-Poissonisation then gives $b_n = \Psi(n) (1 + O(1/n))$, and (11) yields $\Psi(n) = \Phi(\log n) (1 + O(e^{-cn}))$. Hence

$$b_{n,p} = \Phi_p(\log n) (1 + O(1/n)), \quad (12)$$

with $\Phi_p := \Phi$ continuous, strictly positive and L -periodic. This is Theorem 1.2 apart from the non-constancy of Φ_p .

3.5 Non-constancy

By (12), $b_{n,p}$ converges if and only if Φ_p is constant, equivalently $\mathcal{M}[\Xi](2\pi im/L) = 0$ for every $m \neq 0$. At $p = \frac{1}{2}$ one has $b_n \equiv \frac{1}{2}$ and $\Phi_p \equiv \frac{1}{2}$. For $p < \frac{1}{2}$ the non-constancy of Φ_p — that $b_{n,p}$ genuinely oscillates — is the arithmetic heart of the paper; it is Gap D (Section 7).

Remark 3.2 (the fair point). Near $p = \frac{1}{2}$ the non-constancy is explicit. With $p = \frac{1}{2} - \varepsilon$ and $b_{n,p} = \frac{1}{2} - \varepsilon\beta_n + O(\varepsilon^2)$, the first-order perturbation has generating function $\mathcal{B}(z) = z e^{z/2} + (e^{z/2} - 1) \mathcal{B}(z/2)$; the same product identity gives its profile Fourier coefficients $-L^{-1}\Gamma(1 + s_m) D(1 + s_m)$, with $s_m = 2\pi i m/L$, $L = \log 2$, and $D(\sigma) = \sum_{m \geq 1} (-1)^{s_2(m)} m^{-\sigma}$ the *Thue–Morse Dirichlet series* (s_2 the binary digit-sum). Since $D(1 + 2\pi i/L) = -2.320 - 0.313i \neq 0$ (numerically), the first-order perturbation oscillates, and Φ_p is non-constant for p slightly below $\frac{1}{2}$.

4 Step A: $w_{n,p}$ is not eventually increasing

Proposition 4.1. *For $p \in (0, \frac{1}{2})$ there is no M with $w_n \geq w_{n-1}$ for all $n > M$. That is, $w_n < w_{n-1}$ for infinitely many n .*

Proof. Suppose $w_n \geq w_{n-1}$ for all $n > M$, i.e. w is non-decreasing on $\{M, M+1, \dots\}$. Fix $n > M+1$ and split the Bellman sum (1) at $j = M$, writing $S_{j,n} := \max_{j \leq m \leq n-1} w_m$. For $j \geq M$ the index window lies in the non-decreasing range, so $S_{j,n} = w_{n-1}$; for every j one has $w_{n-1} \leq S_{j,n} \leq 1$. With $U_n := \sum_{j=1}^{M-1} \binom{n}{j} p^{n-j} q^j$ and $T_n := \sum_{j=1}^{M-1} \binom{n}{j} p^{n-j} q^j S_{j,n}$,

$$w_n = p^n + T_n + (1 - p^n - q^n - U_n) w_{n-1},$$

hence

$$w_n - w_{n-1} = p^n(1 - w_{n-1}) - q^n w_{n-1} + (T_n - U_n w_{n-1}).$$

Since $w_{n-1} \leq S_{j,n} \leq 1$ one has $0 \leq T_n - U_n w_{n-1} \leq U_n$, and $U_n \leq (M-1) n^{M-1} p^{n-M+1} = C_1 n^{M-1} p^n$ with $C_1 = (M-1)p^{1-M}$. With $c := w_M > 0$ and $w_{n-1} \geq c$,

$$w_n - w_{n-1} \leq (1 + C_1 n^{M-1}) p^n - c q^n.$$

As $p/q < 1$, the ratio $(1 + C_1 n^{M-1})(p/q)^n \rightarrow 0$, so the right-hand side is negative for all large n — contradicting $w_n \geq w_{n-1}$. $\square$

Proposition 4.1 is statement (A) of Section 1.3 and is fully rigorous.

5 Step B, part 1: reduction and the transfer lemma

Lemma 5.1 (reduction). *A bounded real sequence (x_n) that is neither eventually non-decreasing nor eventually non-increasing has infinitely many strict local maxima and infinitely many strict local minima, provided $x_n \neq x_{n+1}$ for all large n .*

Proof. Not eventually non-decreasing means $d_n := x_{n+1} - x_n < 0$ infinitely often; not eventually non-increasing means $d_n > 0$ infinitely often. Between a sign change of d from $-$ to $+$ lies a local minimum, between $+$ and $-$ a local maximum; infinitely many sign changes yield infinitely many of each, strict under $x_n \neq x_{n+1}$. $\square$

Lemma 5.2 (transfer to strategy ALL). *Suppose $w_n \geq w_{n+1}$ for all $n \geq M$. Define $x_0 := 1$, $x_j := \max_{j \leq m \leq M} w_m$ for $1 \leq j \leq M-1$, and $x_j := w_j$ for $j \geq M$. Then $x_n = w_n$ for $n \geq M$, and (x_n) satisfies the exact strategy-ALL recursion*

$$x_n = \sum_{j=0}^{n-1} \binom{n}{j} p^{n-j} q^j x_j \quad \text{for every } n > M.$$

Proof. Fix $n > M$. In the Bellman equation (1) the suffix-maximum $S_{j,n} = \max_{j \leq m \leq n-1} w_m$ collapses to the *left* endpoint on the non-increasing range: for $M \leq j \leq n-1$, $S_{j,n} = w_j = x_j$; for $1 \leq j \leq M-1$, $S_{j,n} = \max(\max_{j \leq m \leq M-1} w_m, w_M) = \max_{j \leq m \leq M} w_m = x_j$, independent of n . Writing the p^n term as $\binom{n}{0} p^n q^0 x_0$ gives the displayed recursion. $\square$

Lemma 5.2 is the crux of Step B: it says that a non-increasing tail of w is, exactly, a strategy-ALL sequence with finitely many initial values modified. The remaining task is to show such a sequence cannot exist, by transferring the oscillation of Theorem 1.2.

6 Step B, part 2: the renormalisation argument

Throughout this section (x_n) is a bounded solution of the strategy-ALL recursion for $n > M$ — either b itself, or a transferred sequence of Lemma 5.2.

6.1 The recursion as a smoothing in log-scale

With $J_n \sim \text{Bin}(n, q)$ the recursion reads $x_n = E[x_{J_n}]/(1 + q^n)$. Read on the log-scale $t := \log n$, and using $\log(qn) = \log n - L$, one application of the recursion shifts the scale down by exactly one period L and smooths by a kernel of width $\sigma(t) \sim \sqrt{p/q} e^{-t/2} \rightarrow 0$.

6.2 The local fundamental coefficient

Fix a window $\chi \in C_c^\infty(\mathbb{R})$ with $\int \chi = 1$ and $\hat{\chi}(2\pi) = 0$, and set

$$A(t) := \frac{1}{L} \sum_{n > M} x_n e^{-i\omega \log n} \chi\left(\frac{\log n - t}{L}\right) \frac{1}{n}. \quad (13)$$

This is a windowed Fourier sum at frequency ω on the multiplicative grid, the weight $1/n$ being the log-scale element $d(\log n)$. The condition $\hat{\chi}(2\pi) = 0$ makes A annihilate the constant (mean-level) component, so A extracts the fundamental oscillation. If x_n converges then $A(t) \rightarrow 0$.

6.3 The propagation estimate

Proposition 6.1 (propagation). *There are $\kappa(t) \in \mathbb{C}$ and a remainder $r(t)$ with*

$$A(t+L) = \kappa(t) A(t) + r(t), \quad |\kappa(t) - 1| = O(e^{-t}), \quad |r(t)| = O(e^{-t}).$$

The factor κ is fully under control.

Lemma 6.2 (the factor κ). *For $J_n \sim \text{Bin}(n, q)$, $\kappa(n) = E[(J_n/(qn))^{i\omega} \mathbf{1}_{J_n \geq 1}] = 1 - \frac{p}{2qn}(\omega^2 + i\omega) + O(n^{-3/2})$. In particular $|\kappa - 1| = O(1/n) = O(e^{-t})$.*

Proof (sketch). Put $V_n = J_n/(qn) - 1$; the binomial central moments give $E[V_n^2] = p/(qn)$ and $E[|V_n|^3] = O(n^{-3/2})$. Expanding $(1 + V_n)^{i\omega}$ to second order on $\{|V_n| \leq \frac{1}{2}\}$ and bounding the tail by Hoeffding yields the claim. $\square$

For the remainder we pass to the *transposed* kernel. This makes the negative-binomial structure of the recursion explicit and, as it turns out, removes the need for any limit theorem. Write $a_n(t) := n^{-1} e^{-i\omega \log n} \chi\left(\frac{\log n - t}{L}\right)$, so that $A(t) = \frac{1}{L} \sum_{n > M} x_n a_n(t)$.

Lemma 6.3 (transposed kernel and the negative binomial). *Substituting the recursion $x_n = (1 + q^n)^{-1} \sum_{j=0}^n P(\text{Bin}(n, q) = j) x_j$ into $A(t + L)$ and exchanging the (finite) order of summation,*

$$A(t + L) = \frac{1}{L} \sum_j x_j W(j, t), \quad W(j, t) = \sum_{n \geq j} \frac{P(\text{Bin}(n, q) = j)}{1 + q^n} a_n(t + L).$$

The transposed weight has exact total mass

$$\sum_{n \geq j} P(\text{Bin}(n, q) = j) = q^j \sum_{k \geq 0} \binom{j+k}{j} p^k = q^j (1-p)^{-(j+1)} = \frac{1}{q},$$

and $q P(\text{Bin}(n, q) = j)$ is the law of $N_j := j + \text{NB}(j+1, q)$, with

$$E[N_j] = \bar{n}_j := \frac{j+p}{q}, \quad \text{Var}(N_j) = \frac{(j+1)p}{q^2}.$$

Consequently $W(j, t) = \frac{1}{q} E[(1 + q^{N_j})^{-1} a_{N_j}(t + L)]$.

Proof. The substitution is valid because the window lies far above M ; the swap is exact since χ has compact support, so every sum is finite. The indices $j \leq M$ contribute super-exponentially little ($P(\text{Bin}(n, q) = j)$ is super-exponentially small for fixed j and n in the window) and are absorbed silently. The mass identity is the negative-binomial series $\sum_{k \geq 0} \binom{j+k}{k} p^k = (1-p)^{-(j+1)}$; hence $q^{j+1} \binom{j+k}{k} p^k$, $k \geq 0$, is the law of $\text{NB}(j+1, q)$ in $k = n - j$, and its mean and variance, shifted by j , are those stated. $\square$

The exact identity is the crux. The transposed weight carries mass *exactly* $1/q$ and is a genuine probability law — being a shifted sum of i.i.d. geometrics, an exceptionally well-behaved one. The oscillatory cancellation is delivered by the recursion's own combinatorial structure; no local limit theorem enters.

Proof of Proposition 6.1. The transposed picture settles κ and r together: we show $A(t + L) = A(t) + O(e^{-t})$, and then $r(t) = A(t + L) - \kappa(t)A(t) = (1 - \kappa(t))A(t) + O(e^{-t}) = O(e^{-t})$, using $|\kappa - 1| = O(e^{-t})$ (Lemma 6.2) and the boundedness of A .

For real $n > 0$ the map $n \mapsto a_n(t + L)$ is smooth, supported in the geometric window $\log n \in t + L + L \text{ supp } \chi$, and every derivative obeys $|a_n^{(k)}(t + L)| \leq C_k n^{-k-1}$: each n -derivative produces one factor n^{-1} and bounded constants $(\omega, L^{-1}, \|\chi^{(k)}\|_\infty)$. The factor $(1 + q^n)^{-1} = 1 + O(q^n)$ perturbs this super-exponentially little.

Fix j with $\bar{n}_j$ within $O(1)$ of the window, so j is of order e^t . A second-order Taylor expansion of $a_n(t + L)$ about $n = \bar{n}_j$, taken under $E[\cdot]$, loses its first-order term because $E[N_j - \bar{n}_j] = 0$:

$$E[a_{N_j}(t + L)] = a_{\bar{n}_j}(t + L) + \frac{1}{2} E[a_{\xi_j}''(t + L) (N_j - \bar{n}_j)^2],$$

with ξ_j between N_j and $\bar{n}_j$, hence $\xi_j \geq j$ (both $N_j \geq j$ and $\bar{n}_j > j$). Thus $|a_{\xi_j}''| \leq C_2 j^{-3}$ and, since $\text{Var}(N_j) = O(j)$,

$$|E[a_{N_j}(t + L)] - a_{\bar{n}_j}(t + L)| \leq \frac{1}{2} C_2 j^{-3} \text{Var}(N_j) = O(j^{-2}).$$

With the super-exponentially small $(1 + q^{N_j})^{-1}$ correction, $W(j, t) = \frac{1}{q} a_{\bar{n}_j}(t + L) + O(j^{-2})$.

It remains to undo the period shift. From $\bar{n}_j = (j + p)/q$,

$$\log \bar{n}_j = \log j + L + \log(1 + p/j), \quad e^{-i\omega L} = 1,$$

and $\log(1 + p/j) = O(j^{-1})$; expanding the three factors of $a_{\bar{n}_j}(t + L)$ gives $\frac{1}{q} a_{\bar{n}_j}(t + L) = a_j(t) (1 + O(j^{-1})) = a_j(t) + O(j^{-2})$, the last step because $a_j(t) = O(j^{-1})$. Hence $W(j, t) = a_j(t) + O(j^{-2})$ uniformly in j , and

$$A(t + L) - A(t) = \frac{1}{L} \sum_j x_j [W(j, t) - a_j(t)], \quad |A(t + L) - A(t)| \leq \frac{\|x\|_\infty}{L} \sum_j O(j^{-2}).$$

The window confines j to $[e^{t-cL}, e^{t+cL}]$ with c fixed by $\text{supp } \chi$; all these indices exceed e^{t-cL} , so $\sum_j O(j^{-2}) = O(e^{-t})$. This proves $A(t + L) = A(t) + O(e^{-t})$, and with it Proposition 6.1. $\square$

In its earlier form the remainder bound was a separate gap — a local central limit theorem, with one Edgeworth correction, for the binomial log-deviation density. The transposed picture closes it: the cancellation is supplied by the *exact* negative-binomial identity of Lemma 6.3, and the windowed coefficient obeys the elementary propagation law $A(t + L) = A(t) + O(e^{-t})$.

6.4 From propagation to non-convergence

Theorem 6.4 (renormalisation). *Let (x_n) be a bounded solution of the strategy-ALL recursion for $n > M$. Then for each t_0 the sequence $A(t_0 + kL)$ converges, as $k \rightarrow \infty$, to a limit A_∞ . Moreover: if x_n converges then $A_\infty = 0$; and if $A_\infty \neq 0$ then (x_n) is not eventually monotone.*

Proof. Write $A_k = A(t_0 + kL)$, $\kappa_k = \kappa(t_0 + kL)$, $r_k = r(t_0 + kL)$. By Proposition 6.1, $\sum_k |\kappa_k - 1| < \infty$ and $\sum_k |r_k| < \infty$ (geometric series in $e^{-L} < 1$). Solving $A_{k+1} = \kappa_k A_k + r_k$,

$$A_k = \left(\prod_{j < k} \kappa_j \right) A_0 + \sum_{j < k} \left(\prod_{j < i < k} \kappa_i \right) r_j \rightarrow A_\infty,$$

the products and series converging absolutely. If x_n converges then $A(t) \rightarrow 0$, so $A_\infty = 0$. If $A_\infty \neq 0$ then $A_k \not\rightarrow 0$, so x_n does not converge; a bounded non-convergent sequence is not eventually monotone. $\square$

7 The base case and the proof of Theorem 1.1

Theorem 6.4 reduces Step B to a single non-degeneracy: the limit A_∞ must be non-zero.

Proposition 7.1 (base case). *For every transferred sequence (x_n) of Lemma 5.2, $A_\infty \neq 0$.*

By linearity $A_\infty(x) = A_\infty(b) + A_\infty(g)$, where $g := x - b$ satisfies $g \geq 0$ (the optimum dominates strategy ALL) and solves the strategy-ALL recursion for $n > M$.

► OPEN GAP — Gap D — the non-degeneracy. Work continues here.

$A_\infty(x) \neq 0$ is the genuine arithmetic heart of the proof. It needs two things: that pure strategy ALL oscillates — $A_\infty(b) \neq 0$, the non-constancy of Φ_p (Section 3.5) — and that the perturbation g does not cancel that oscillation, $A_\infty(g) \neq -A_\infty(b)$. The first is settled near $p = \frac{1}{2}$ (Section 3.5) and expected throughout $(0, \frac{1}{2})$; the second is not secured by $g \geq 0$ alone, and must draw on the optimality of w . Work continues here.

Proof of Theorem 1.1. Fix $p \in (0, \frac{1}{2})$. By Lemma 5.1 it suffices to show that $(w_{n,p})$ is neither eventually non-decreasing nor eventually non-increasing.

Not eventually non-decreasing is Proposition 4.1.

Not eventually non-increasing. Suppose, for contradiction, $w_n \geq w_{n+1}$ for all $n \geq M$. By Lemma 5.2 the transferred sequence (x_n) satisfies $x_n = w_n$ for $n \geq M$ and solves the strategy-ALL recursion for $n > M$; being a tail of a monotone bounded sequence, x_n converges. By Theorem 6.4 then $A_\infty = 0$, contradicting Proposition 7.1. Hence no non-increasing tail exists.

Both one-sided statements hold, so $(w_{n,p})$ is not eventually monotone; Lemma 5.1 gives infinitely many strict local maxima and minima. $\square$

Remark 7.2 (strictness). Lemma 5.1 delivers *strict* extrema once $w_{n,p} \neq w_{n+1,p}$ for all large n , which holds in all computed data and is expected for every $p \in (0, \frac{1}{2})$; absent it, Theorem 1.1 holds with the non-strict notion of local extremum.

8 Numerical evidence

The two predictions have been checked numerically in arbitrary-precision arithmetic (`simulation/b.asymptotics.py`, `simulation/renorm_validation.py`). For strategy ALL the log-periodic oscillation of Theorem 1.2 is confirmed across fourteen orders of magnitude in its amplitude. For the renormalisation argument: the windowed coefficient $A(t_0 + kL)$ is observed to converge to a non-zero limit A_∞ for every tested $p \in \{0.35, 0.40, 0.42, 0.45\}$, and $n|\kappa(n) - 1|$ converges to the constant $\frac{p}{2q}|\omega^2 + i\omega|$ of Lemma 6.2 — a direct quantitative confirmation of Proposition 6.1. The non-constancy of Φ_p near $p = \frac{1}{2}$ is confirmed separately (`simulation/d1_perturbation.py`, `d1_thue_morse.py`): the first-order perturbation oscillates, and the Thue–Morse value $D(1 + 2\pi i/L)$ is non-zero.

9 Status and open problems

The skeleton of the proof is complete and the rigorous backbone is now substantial: the reduction (Lemma 5.1), the elementary Step A (Proposition 4.1), the exact transfer lemma (Lemma 5.2), the log-periodic asymptotics of strategy ALL (Section 3: the product identity (9) and de-Poissonisation), the factor estimate (Lemma 6.2), the propagation estimate (Proposition 6.1, via the transposed negative-binomial kernel), and the renormalisation theorem (Theorem 6.4). A single gap remains open; it is flagged in the text and summarised here.

Gap	What is missing	Where
D	the non-degeneracy $A_\infty(x) \neq 0$ (transferred sequence)	§7

A complete proof of Theorem 1.1 now consists of Proposition 4.1 (done) together with Gap D alone. Three gaps of earlier drafts have closed. Gap A (a Hayman-type coefficient transfer): the strategy-ALL asymptotics are established in Section 3 by the exact product identity and analytic de-Poissonisation, with no admissibility theory. Gap B (a Mellin contour shift): the same route makes it unnecessary. Gap C (the propagation remainder): closed by the transposed kernel (Lemma 6.3) via the exact identity $\sum_{n \geq j} P(\text{Bin}(n, q) = j) = 1/q$. The sole remaining problem is the non-degeneracy $A_\infty(x) \neq 0$ (Gap D), the arithmetic heart: that strategy ALL oscillates — settled near $p = \frac{1}{2}$ (Section 3.5) — and that the optimal perturbation does not cancel that oscillation. This is the current front of work.

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